

3^k FD with two factors

Kontadakis Eftychis

Introduction

In this work we are going to apply a 3^k FD in a 2 factor experiment.

Data

We have 2 replicates in each combination level.

```
# Factors
A <- factor(rep(c("A1","A2","A3"), each = 6))
B <- factor(rep(rep(c("B1","B2","B3"), each = 2), times = 3))

Response <- c(-2, -1, # A1-B1, 2 replicates
              -3, 0, # A1-B2
               2, 3, # A1-B3
               0, 2, # A2-B1
               1, 3, # A2-B2
               4, 6, # A2-B3
              -1, 0, # A3-B1
               5, 6, # A3-B2
               0, -1) # A3-B3

# Create the dataframe
data <- data.frame(A, B, Response)
```

With the dataframe ready we can proceed to running the actual ANOVA and Regression

ANOVA

```
anova_model <- aov(Response ~ A * B, data = data)
summary(anova_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
A	2	24.33	12.167	8.423	0.00868 **
B	2	25.33	12.667	8.769	0.00770 **
A:B	4	61.33	15.333	10.615	0.00184 **
Residuals	9	13.00	1.444		

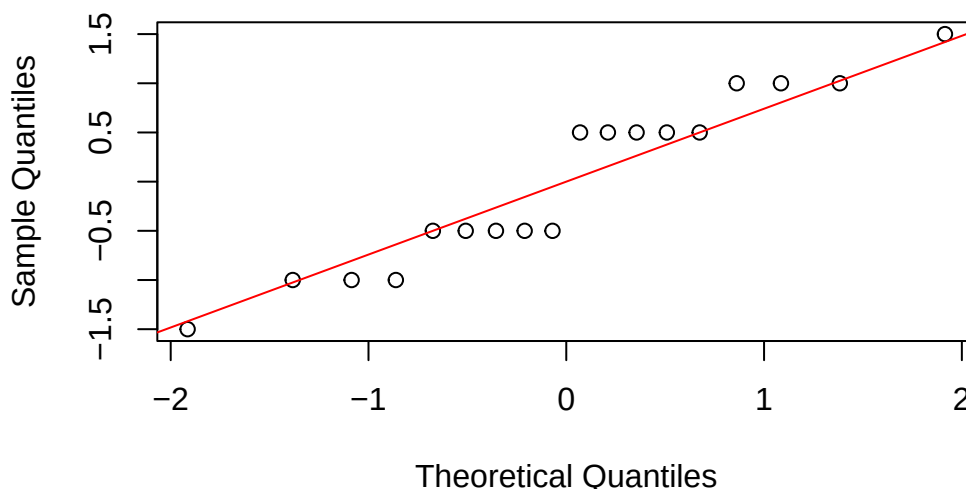
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

We can see that all the effects A,B and AB are important.

We are of course going to show the Q-Q plot to make sure our original assumptions were correct:

```
# QQ plot
qqnorm(residuals(anova_model))
qqline(residuals(anova_model), col = "red")
```

Normal Q-Q Plot



Regression

```
# Set the orthogonal contrasts:
contrasts(data$A) <- contr.poly(3)
contrasts(data$B) <- contr.poly(3)

reg_model <- lm(Response ~ A * B, data = data)
summary(reg_model)
```

Call:

```
lm(formula = Response ~ A * B, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.5	-0.5	0.0	0.5	1.5

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.3333	0.2833	4.707	0.001110	**
A.L	1.1785	0.4907	2.402	0.039772	*
A.Q	-1.6330	0.4907	-3.328	0.008824	**
B.L	1.8856	0.4907	3.843	0.003948	**
B.Q	-0.8165	0.4907	-1.664	0.130451	
A.L:B.L	-2.0000	0.8498	-2.353	0.043065	*
A.Q:B.L	-1.1547	0.8498	-1.359	0.207306	
A.L:B.Q	-4.6188	0.8498	-5.435	0.000414	***
A.Q:B.Q	-2.0000	0.8498	-2.353	0.043065	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

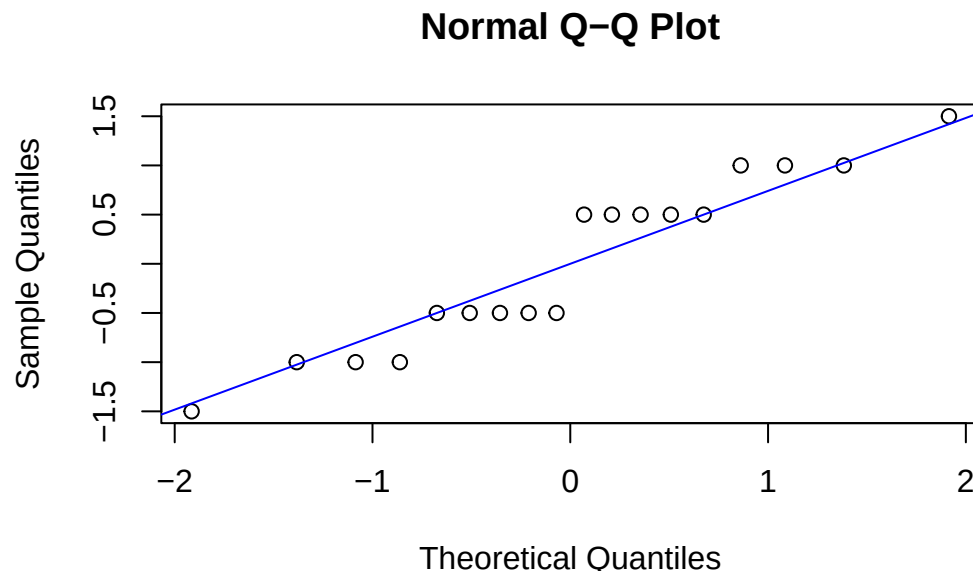
Residual standard error: 1.202 on 9 degrees of freedom

Multiple R-squared: 0.8952, Adjusted R-squared: 0.802

F-statistic: 9.606 on 8 and 9 DF, p-value: 0.001337

Again the Q-Q Plot:

```
# QQ plot
qqnorm(residuals(reg_model))
qqline(residuals(reg_model), col = "blue")
```



As expected the Q-Q plots much completely for both the ANOVA and the Regression. The results also match the expected that were given in class.